Example 21.8. *Find* $L\left\{\frac{\sin at}{t}\right\}$, *given that* $L\left\{\frac{\sin t}{t}\right\} = \tan^{-1}\left\{\frac{1}{s}\right\}$.

Solution. By the above property,

$$L\left\{\frac{\sin at}{at}\right\} = \frac{1}{a}\tan^{-1}\left\{\frac{1}{(s/a)}\right\} = \frac{1}{a}\tan^{-1}\left(\frac{a}{s}\right) \text{ i.e., } L\left\{\frac{\sin at}{t}\right\} = \tan^{-1}\left\{\frac{a}{s}\right\}.$$

PROBLEMS 21.1

Find the Laplace transforms of

1. $e^{2t} + 4t^3 - 2\sin 3t + 3\cos 3t$. (*J.N.T.U., 2003*)
2. $1 + 2\sqrt{t} + 3/\sqrt{t}$.
3. $3\cosh 5t - 4\sinh 5t$. (*Nagarjuna, 2006*)
4. $\cos(at + b)$.
5. $(\sin t - \cos t)^2$.
6. $\sin 2t \cos 3t$. (*Kottayam, 2005*)
7. $\sin\sqrt{t}$.
8. $\sin^5 t$. (*Mumbai, 2007*)
9. $\cos^3 2t$.
10. $e^{-at}\sinh bt$
11. $e^{2t}(3t^5 - \cos 4t)$. (*P.T.U., 2007*)
12. $e^{-3t}\sin 5t \sin 3t$. (*V.T.U., 2006*)
13. $e^{-t}\sin^2 t$. (*Mumbai, 2009*)
14. $e^{2t}\sin^4 t$ (*Mumbai, 2007*)
15. $\cosh at \sin at$. (*Delhi, 2002*)
16. $\sinh 3t \cos^2 t$ (*Madras, 2000*)
17. $t^2 e^{2t}$. (*V.T.U., 2008 S*)
18. $(1 + te^{-t})^3$
19. $t\sqrt{(1 + \sin t)}$ (*Mumbai, 2007*)
20. $f(t) = \begin{cases} 4, & 0 \le t <= 1 \\ 3, & t > 1 \end{cases}$ (*U.P.T.U., 2009*)
21. $f(t) = \begin{cases} \sin t, 0 < t < \pi \\ 0, t > \pi. \end{cases}$ (*Madras, 2000 S*)
22. $f(x) = \begin{cases} \sin(x - \pi/3), & x > \pi/3 \\ 0, & x < \pi/3 \end{cases}$ (*Rajasthan, 2006*)
23. $f(t) = \begin{cases} \cos(t - 2\pi/3), & t > 2\pi/3 \\ 0, & t < 2\pi/3 \end{cases}$
24. $f(t) = \begin{cases} t^2, 0 < t < 2 \\ t - 1, 2 < t < 3 \\ 7, t > 3. \end{cases}$ (*Mumbai, 2007*)
25. If $L[f(t)] = \frac{1}{s(s^2+1)}$, find $L[e^{-t}f(2t)]$.

21.5 TRANSFORMS OF PERIODIC FUNCTIONS

If $f(t)$ is a periodic function with period T, i.e., $f(t + T) = f(t)$, then

$$L\{f(t)\} = \frac{\int_0^T e^{-st} f(t)dt}{1 - e^{-sT}}$$

We have $$L\{f(t)\} = \int_0^\infty e^{-st} f(t)dt = \int_0^T e^{-st} f(t)dt + \int_T^{2T} e^{-st} f(t)dt + \int_{2T}^{3T} e^{-st} f(t)dt + \ldots$$

In the second integral put $t = u + T$, in the third integral put $t = u + 2T$, and so on. Then

$$L\{f(t)\} = \int_0^T e^{-st} f(t)dt + \int_0^T e^{-s(u+T)} f(u+T)du + \int_0^T e^{-s(u+2T)} f(u+2T)du + \ldots$$

$$= \int_0^T e^{-st} f(t)dt + e^{-sT}\int_0^T e^{-su} f(u)du + e^{-2sT}\int_0^T e^{-su} f(u)du + \ldots$$

$$[\because f(u) = f(u+T) = f(u+2T) \text{ etc.}]$$

$$= \int_0^T e^{-st} f(t)dt + e^{-sT}\int_0^T e^{-st} f(t)dt + e^{-2sT}\int_0^T e^{-st} f(t)dt + \ldots$$

$$= (1 + e^{-sT} + e^{-2sT} + \ldots \infty)\int_0^T e^{-st} f(t)dt$$

$$= \frac{1}{1 - e^{-sT}}\int_0^T e^{-st} f(t)dt.$$ (*V.T.U., 2008 ; Mumbai, 2006*)

Example 21.9. *Find the Laplace transform of the function*

$$f(t) = \sin \omega t, \quad 0 < t < \pi/\omega$$
$$= 0, \quad \pi/\omega < t < 2\pi/\omega.$$

(*Kurukshetra, 2005 ; Madras, 2003*)

Solution. Since $f(t)$ is a periodic function with period $2\pi/\omega$.

$$\therefore \quad L\{f(t)\} = \frac{1}{1-e^{-2\pi s/\omega}} \int_0^{2\pi/\omega} e^{-st} f(t) dt$$

$$= \frac{1}{1-e^{-2\pi s/\omega}} \left[\int_0^{\pi/\omega} e^{-st} \sin \omega t \, dt + \int_{\pi/\omega}^{2\pi/\omega} e^{-st} . 0 \, dt \right]$$

$$= \frac{1}{1-e^{-2\pi s/\omega}} \left| \frac{e^{-st}(-s \sin \omega t - \omega \cos \omega t)}{s^2+\omega^2} \right|_0^{\pi/\omega} = \frac{\omega e^{-\pi s/\omega} + \omega}{(1-e^{-2\pi s/\omega})(s^2+\omega^2)} = \frac{\omega}{(1-e^{-\pi s/\omega})(s^2+\omega^2)}.$$

Example 21.10. *Draw the graph of the periodic function*

$$f(t) = \begin{cases} t, & 0 < t < \pi \\ \pi - t, & \pi < t < 2\pi. \end{cases}$$

and find its Laplace transform. (*U.P.T.U., 2003*)

Solution. Here the period of $f(t) = 2\pi$ and its graph is as in Fig. 21.1.

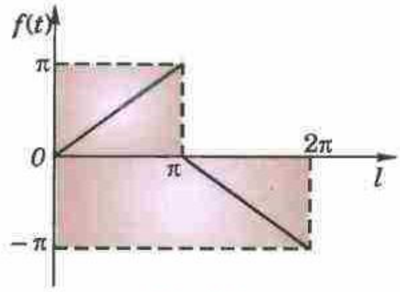

Fig. 21.1

$$\therefore \quad L f(t) = \frac{1}{1-e^{-2\pi s}} \left\{ \int_0^{\pi} e^{-st} t \, dt + \int_{\pi}^{2\pi} e^{-st} (\pi - t) dt \right\}$$

$$= \frac{1}{1-e^{-2\pi s}} \left\{ \left| t\left(\frac{e^{-st}}{-s}\right) - 1 . \left(\frac{e^{-st}}{s^2}\right) \right|_0^{\pi} + \left| (\pi - t)\left(\frac{e^{-st}}{-s}\right) - (-1)\left(\frac{e^{-st}}{s^2}\right) \right|_{\pi}^{2\pi} \right\}$$

$$= \frac{1}{1-e^{-2\pi s}} \left\{ \frac{-\pi e^{-\pi s}}{s} - \frac{e^{-\pi s}}{s^2} + \frac{1}{s^2} + \frac{\pi e^{-2\pi s}}{s} + \frac{e^{-2\pi s}}{s^2} - \frac{e^{-\pi s}}{s^2} \right\}$$

$$= \frac{1}{1-e^{-2\pi s}} \left\{ \frac{\pi}{s}\left(e^{-2\pi s} - e^{-\pi s}\right) + \frac{1}{s^2}\left(1 + e^{-2\pi s} - 2e^{-\pi s}\right) \right\}.$$

21.6 TRANSFORMS OF SPECIAL FUNCTIONS

(1) Transform of Bessel functions $J_0(x)$ and $J_1(x)$.

We know that $$J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 . 4^2} - \frac{x^6}{2^2 . 4^2 . 6^2} + \ldots$$ [§ 16.7 (1), p. 553]

$$\therefore \quad L\{J_0(x)\} = \frac{1}{s} - \frac{1}{2^2}\frac{2!}{s^3} + \frac{1}{2^2 . 4^2} . \frac{4!}{s^5} - \frac{1}{2^2 . 4^2 . 6^2} . \frac{6!}{s^7} + \ldots$$

$$= \frac{1}{s}\left\{1 - \frac{1}{2}\left(\frac{1}{s^2}\right) + \frac{1 . 3}{2 . 4}\left(\frac{1}{s^4}\right) - \frac{1 . 3 . 5}{2 . 4 . 6}\left(\frac{1}{s^6}\right) + \ldots\right\}$$

$$= \frac{1}{s}\left(1 + \frac{1}{s^2}\right)^{-1/2} = \frac{1}{\sqrt{(s^2+1)}} \quad \ldots(1)$$

Also since $$J_0'(x) = -J_1(x).$$ [Problem 4(*i*), p. 557]

$$\therefore \quad L\{J_1(x)\} = -L\{J_0'(x)\} = -[sL\{J_0(x)\} - 1] = 1 - \frac{s}{\sqrt{(s^2+1)}} \quad \ldots(2)$$

(2) Transform of Error function

We know that $$erf(\sqrt{x}) = \frac{2}{\sqrt{\pi}} \int_0^{\sqrt{x}} e^{-t^2} dt$$ (§ 7.18, p. 312)

$$= \frac{2}{\sqrt{\pi}} \int_0^{\sqrt{x}} \left(1 - t^2 + \frac{t^4}{2!} - \frac{t^6}{3!} + \ldots\right) dt = \frac{2}{\sqrt{\pi}} \left(x^{1/2} - \frac{x^{3/2}}{3} + \frac{x^{5/2}}{5.2!} - \frac{x^{7/2}}{7.3!} + \ldots\right)$$

$$\therefore \quad L\{erf(\sqrt{x})\} = \frac{2}{\sqrt{\pi}} \left\{\frac{\Gamma(3/2)}{s^{3/2}} - \frac{\Gamma(5/2)}{3s^{5/2}} + \frac{\Gamma(7/2)}{5.2!\,s^{7/2}} - \frac{\Gamma(9/2)}{7.3!\,s^{9/2}} + \ldots\right\}$$

$$= \frac{1}{s^{3/2}} - \frac{1}{2}\frac{1}{s^{5/2}} + \frac{1.3}{2.4} \cdot \frac{1}{s^{7/2}} - \frac{1.3.5}{2.4.6} \cdot \frac{1}{s^{9/2}} + \ldots$$

$$= \frac{1}{s^{3/2}} \left\{1 - \frac{1}{2} \cdot \frac{1}{s} + \frac{1.3}{2.4} \cdot \frac{1}{s^2} - \frac{1.3.5}{2.4.6} \cdot \frac{1}{s^3} + \ldots\right\}$$

$$= \frac{1}{s^{3/2}} \left[1 + \frac{1}{s}\right]^{-1/2} = \frac{1}{s\sqrt{(s+1)}}. \qquad \textit{(Mumbai, 2009)} \quad \ldots(3)$$

(3) Transform of Laguerre's polynomials $L_n(x)$

We know that $L_n(x) = e^x \frac{d^n}{dx^n}(x^n e^{-x})$ (§ 16.18, p. 571)

$$L[L_n(t)] = \int_0^\infty e^{-st} e^t \frac{d^n}{dt^n}(t^n e^{-t}) dt = \int_0^\infty e^{-(s-1)t} \frac{d^n}{dt^n}(e^{-t} t^n) dt$$

$$= \left| e^{-(s-1)t} \frac{d^{n-1}}{dt^{n-1}}(e^{-t} t^n) \right|_0^\infty + \int_0^\infty e^{-(s-1)t\,(s-1)}\,(s-1) \frac{d^{n-1}}{dt^{n-1}}(e^{-t} t^n) dt$$

$$= (s-1) \int_0^\infty e^{-(s-1)t} \frac{d^{n-1}}{dt^{n-1}}(e^{-t} t^n) dt. \qquad \text{(Integrating by parts)}$$

$$= (s-1)^n \int_0^\infty e^{-(s-1)t} \,.\, e^{-t} \,.\, t^n\, dt = (s-1)^n \int_0^\infty e^{-st} \,.\, t^n\, dt$$

$$= (s-1)^n L(t^n) = (s-1)^n \,.\, \frac{n!}{s^{n+1}}$$

Hence $$L[L_n(x)] = \frac{n!(s-1)^n}{s^n + 1} \quad (s > 1).$$

Example 21.11. *Evaluate (i) $L\{e^{-at} J_0(at)\}$ (ii) $L(erf\, 2\sqrt{t}\,)$.* *(Mumbai, 2006)*

Solution. (*i*) We know that $L\{J_0(at)\} = \frac{1}{\sqrt{(s^2 + a^2)}}$

By shifting property, we get

$$L\{e^{-at} J_0(at)\} = \frac{1}{\sqrt{[(s+a)^2 + a^2]}} = \frac{1}{\sqrt{(s^2 + 2sa + 2a^2)}}$$

(*ii*) We know that $L(erf\sqrt{t}\,) = \frac{1}{s(s+1)}$

$$\therefore \quad L(erf\; 2\sqrt{t}\,) = L[erf\sqrt{(4t)}] = \frac{1}{4} \cdot \frac{1}{\frac{s}{4}\sqrt{\left(\frac{s}{4} + 1\right)}} = \frac{2}{s\sqrt{(s+4)}}.$$

PROBLEMS 21.2

1. Find the Laplace transform of the *saw-toothed wave* of period T, given $f(t) = t/T$ for $0 < t < T$. *(V.T.U., 2007)*
2. Find the Laplace transform of the *full-wave rectifier*
$$f(t) = E \sin wt,\ 0 < t < \pi/w, \text{ having period } \pi/w.$$

3. Find the Laplace transform of the *square-wave* (or *meander*) function of period a defined as

$$f(t) = k, \quad \text{when } 0 < t < \alpha$$
$$= -k, \quad \text{when } \alpha < t < 2\alpha.$$ (V.T.U., 2011)

4. Find the Laplace transform of the *triangular wave* of period $2a$ given by

$$f(t) = t, \quad 0 < t < a$$
$$= 2a - t, \quad a < t < 2a.$$ (Nagarjuna, 2008 ; V.T.U., 2008 S ; U.P.T.U., 2002)

Find the Laplace transform of the following functions :

5. $J_0(ax)$. 6. $e^{-at} J_0(bt)$. 7. $e^{2t} erf(\sqrt{t})$.

21.7 TRANSFORMS OF DERIVATIVES

(1) *If $f'(t)$ be continuous and $L\{f(t)\} = \bar{f}(s)$, then $L\{f'(t)\} = s\bar{f}(s) - f(0)$.*

$$L\{f'(t)\} = \int_0^\infty e^{-st} f'(t)dt$$ [Integrate by parts]

$$= \left| e^{-st} f(t) \right|_0^\infty - \int_0^\infty (-s)e^{-st} . f(t)dt.$$

Now assuming $f(t)$ to be such that $\underset{t\to\infty}{\text{Lt}}\ e^{-st} f(t) = 0$. When this condition is satisfied, $f(t)$ is said to be *exponential order s*.

Thus, $$L\{f'(t)\} = f(0) + s\int_0^\infty e^{-st} f(t)dt$$

whence follows the desired result.

(2) *If $f'(t)$ and its first $(n-1)$ derivatives be continuous, then*

$$\mathbf{L\{f^n(t)\} = s^n \bar{f}(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{n-1}(0).}$$

Using the general rule of integration by parts (Footnote p. 398).

$$L\{f^n(t)\} = \int_0^\infty e^{-st} f^n(t)dt$$
$$= |e^{-st} f^{n-1}(t) - (-s)e^{-st} f^{n-2}(t) + (-s)^2 e^{-st} f^{n-3}(t) - \dots$$
$$+ (-1)^{n-1}(-s)^{n-1} e^{-st} . f(t) |_0^\infty + (-1)^n(-s)^n \int_0^\infty e^{-st} f(t)dt$$
$$= -f^{n-1}(0) - sf^{n-2}(0) - s^2 f^{n-3}(0) - \dots - s^{n-1}f(0) + s^n \int_0^\infty e^{-st} f(t)dt$$

Assuming that $\underset{t\to\infty}{\text{Lt}}\ e^{-st} f^m(t) = 0$ for $m = 0, 1, 2, \dots, n-1$.

This proves the required result.

21.8 TRANSFORMS OF INTEGRALS

If $L\{f(t)\} = \bar{f}(s)$, then $\mathbf{L\left\{\int_0^t f(u)du\right\} = \frac{1}{s}\bar{f}(s).}$

Let $$\phi(t) = \int_0^t f(u)\,du, \text{ then } \phi'(t) = f(t) \text{ and } \phi(0) = 0$$

$\therefore$ $$L\{\phi'(t)\} = s\bar{\phi}(s) - \phi(0)$$ [By § 21.7 (1)]

or $$\bar{\phi}(s) = \frac{1}{s} L\{\phi'(t)\} \quad i.e., \quad L\left(\int_0^t f(u)du\right) = \frac{1}{s}\bar{f}(s).$$

21.9 MULTIPLICATION BY t^n

If $L\{f(t)\} = \bar{f}(s)$, then

$$\mathbf{L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}[\bar{f}(s)]}, \textit{ where } n = 1, 2, 3 \dots$$

We have $\int_0^\infty e^{-st} f(t)\, dt = \bar{f}(s)$.

Differentiating both sides with respect to s, $\frac{d}{ds}\left\{\int_0^\infty e^{-st} f(t)\, dt\right\} = \frac{d}{ds}\{\bar{f}(s)\}$

or By Leibnitz's rule for differentiation under the integral sign (p. 233).

$$\int_0^\infty \frac{\partial}{\partial s}(e^{-st})\, f(t)\, dt = \frac{d}{ds}\{\bar{f}(s)\}$$

or $$\int_0^\infty \{-te^{-st} f(t)\}\, dt = \frac{d}{ds}[\bar{f}(s)] \quad \text{or} \quad \int_0^\infty e^{-st} [tf(t)]\, dt = -\frac{d}{ds}[\bar{f}(s)]$$

which proves the theorem for $n = 1$.

Now assume the theorem to be true for $n = m$ (say), so that

$$\int_0^\infty e^{-st} [t^m f(t)]\, dt = (-1)^m \frac{d^m}{ds^m}[\bar{f}(s)]$$

Then $$\frac{d}{ds}\left[\int_0^\infty e^{-st}\, t^m f(t)\, dt\right] = (-1)^m \frac{d^{m+1}}{ds^{m+1}}[\bar{f}(s)]$$

or By Leibnitz's rule, $\int_0^\infty (-te^{-st})\,.\, t^m f(t)\, dt = (-1)^m \frac{d^{m+1}}{ds^{m+1}}[\bar{f}(s)]$

or $$\int_0^\infty e^{-st} [t^{m+1} f(t)]\, dt = (-1)^{m+1} \frac{d^{m+1}}{ds^{m+1}}[\bar{f}(s)].$$

This shows that, if the theorem, is true for $n = m$, it is also true for $n = m + 1$. But it is true for $n = 1$. Hence it is true for $n = 1 + 1 = 2$, and $n = 2 + 1 = 3$ and so on.

Thus the theorem is true for all positive integral values of n. *(U.P.T.U., 2005)*

Example 21.12. *Find the Laplace transforms of*
(*i*) *$t \cos at$* *(Raipur, 2005)* (*ii*) *$t^2 \sin at$* *(S.V.T.U., 2007)*
(*iii*) *$t^3 e^{-3t}$* *(Kottayam, 2005)* (*iv*) *$te^{-t} \sin 3t$.* *(Kurukshetra, 2005)*

Solution. (*i*) Since $L(\cos at) = s/(s^2 + a^2)$

$$\therefore \quad L(t \cos at) = -\frac{d}{ds}\left(\frac{s}{s^2 + a^2}\right) = -\frac{s^2 + a^2 - s\,.\,2s}{(s^2 + a^2)^2}$$

$$= \frac{s^2 - a^2}{(s^2 + a^2)^2}$$ [*cf.* Example 21.4]

(*ii*) Since $\sin at = \frac{a}{s^2 + a^2}$,

$$\therefore \quad L(t^2 \sin at) = (-1)^2 \frac{d^2}{ds^2}\left(\frac{a}{s^2 + a^2}\right) = \frac{d}{ds}\left\{\frac{-2as}{(s^2 + a^2)^2}\right\} = \frac{2a(3s^2 - a^2)}{(s^2 + a^2)^3}.$$

(*iii*) Since $L(e^{-3t}) = 1/(s + 3)$,

$$\therefore \quad L(t^3 e^{-3t}) = (-1)^3 \frac{d^3}{ds^3}\left(\frac{1}{s + 3}\right) = -\frac{(-1)^3\,.\,3\,!}{(s + 3)^{3+1}} = 6/(s + 3)^4.$$

(*iv*) Since $L(\sin 3t) = \frac{3}{s^2 + 3^2}$, therefore $L(t \sin 3t) = -\frac{d}{ds}\left(\frac{3}{s^2 + 3^2}\right) = \frac{6s}{(s^2 + 9)^2}$

Now using the shifting property (§ 21.4 II), we get

$$L(e^{-t}\, t \sin 3t) = \frac{6(s + 1)}{[(s + 1)^2 + 9]^2} = \frac{6(s + 1)}{(s^2 + 2s + 10)^2}.$$

Example 21.13. *Evaluate (i) $L\{t\,J_0(at)\}$ (ii) $L\{t\,J_1(t)\}$ (iii) $L\{t\,erf\,2\sqrt{t}\}$.*

Solution. (*i*) Since $L\{J_0(at)\} = \dfrac{1}{\sqrt{(s^2+a^2)}}$

$$\therefore \quad L\{tJ_0(at)\} = -\frac{d}{ds}\{L\,J_0(at)\} = -\frac{d}{ds}\frac{1}{\sqrt{(s^2+a^2)}} = \frac{s}{(s^2+a^2)^{3/2}}$$

(*ii*) Since $L\{J_1(t)\} = 1 - \dfrac{s}{\sqrt{(s^2+1)}}$

$$\therefore \quad L\{t\,J_1(t)\} = -\frac{d}{ds}\{L\,J_0(t)\} = -\frac{d}{ds}\left\{1 - \frac{s}{(\sqrt{s^2+1})}\right\} = \frac{1}{(s^2+1)^{3/2}}$$

(*iii*) Since $L(erf\,\sqrt{t}) = \dfrac{1}{s\sqrt{(s+1)}}$

$$\therefore \quad L(erf\,2\sqrt{t}) = L\,erf\,\sqrt{4(t)} = \frac{1}{4}\cdot\frac{1}{\frac{s}{4}\sqrt{\left(\frac{s}{4}+1\right)}} = \frac{2}{s\sqrt{(s+4)}}$$

Thus
$$L(t\,erf\,2\sqrt{t}) = -\frac{d}{ds}\left\{\frac{2}{s\sqrt{(s+4)}}\right\} = -\frac{d}{ds}\left\{\frac{2}{\sqrt{(s^3+4s^2)}}\right\} = \frac{3s+8}{s^2(s+4)^{3/2}}$$

21.10 DIVISION BY t

If $L\{f(t)\} = \bar{f}(s)$, *then* $\mathbf{L}\left\{\frac{1}{t}\mathbf{f(t)}\right\} = \int_s^\infty \bar{\mathbf{f}}(\mathbf{s})\,\mathbf{ds}$ *provided the integral exists.*

We have $\bar{f}(s) = \int_0^\infty e^{-st} f(t)\,dt$

Integrating both sides with respect to s from s to ∞.

$$\int_s^\infty \bar{f}(s)\,ds = \int_s^\infty \left[\int_0^\infty e^{-st} f(t)\,dt\right] ds = \int_0^\infty \int_s^\infty f(t)\,e^{-st}\,ds\,dt$$

[Changing the order of integration]

$$= \int_0^\infty f(t)\left[\int_s^\infty e^{-st}\,ds\right] dt \qquad [\because \; t \text{ is independent of } s]$$

$$= \int_0^\infty f(t)\left|\frac{e^{-st}}{-t}\right|_s^\infty dt = \int_0^\infty e^{-st}\cdot\frac{f(t)}{t}\,dt = L\left\{\frac{1}{t}f(t)\right\}.$$

Example 21.14. *Find the Laplace transform of (i) $(1-e^t)/t$* *(Madras, 2000)*

(ii) $\dfrac{\cos at - \cos bt}{t} + t\sin at$. *(V.T.U., 2010)*

Solution. (*i*) Since $L(1-e^t) = L(1) - L(e^t) = \dfrac{1}{s} - \dfrac{1}{s-1}$

$$\therefore \quad L\left(\frac{1-e^t}{t}\right) = \int_s^\infty \left(\frac{1}{s} - \frac{1}{s-1}\right) ds = \left|\log s - \log(s-1)\right|_0^\infty$$

$$= \left|\log\left(\frac{s}{s-1}\right)\right|_s^\infty = -\log\left[\frac{1}{1-1/s}\right] = \log\left(\frac{s-1}{s}\right)$$

(*ii*) Since $L(\cos at - \cos bt) = \dfrac{s}{s^2+a^2} - \dfrac{s}{s^2+b^2}$ and $L(\sin at) = \dfrac{a}{s^2+a^2}$

$$\therefore \quad L\left(\frac{\cos at - \cos bt}{t}\right) + L(t \sin at) = \int_s^\infty \left(\frac{s}{s^2+a^2} - \frac{s}{s^2+b^2}\right) ds - \frac{d}{ds}\left(\frac{a}{s^2+a^2}\right)$$

$$= \left|\frac{1}{2}\log(s^2+a^2) - \frac{1}{2}\log(s^2+b^2)\right|_s^\infty - a\,\frac{-2s}{(s^2+a^2)^2}$$

$$= \frac{1}{2}\underset{s\to\infty}{\text{Lt}}\ \log\frac{s^2+a^2}{s^2+b^2} - \frac{1}{2}\log\frac{s^2+a^2}{s^2+b^2} + \frac{2as}{(s^2+a^2)^2}$$

$$= \frac{1}{2}\log\left(\frac{1+0}{1+0}\right) - \frac{1}{2}\log\left(\frac{s^2+a^2}{s^2+b^2}\right) + \frac{2as}{(s^2+a^2)^2} = \log\left(\frac{s^2+b^2}{s^2+a^2}\right)^{1/2} + \frac{2as}{(s^2+a^2)^2}$$

$[\because \ \log 1 = 0]$

Example 21.15. *Evaluate (i)* $L\left\{e^{-t}\int_0^t \frac{\sin t}{t}\,dt\right\}$ *(Madras, 2006)*

(ii) $L\left\{t\int_0^t \frac{e^{-t}\sin t}{t}\,dt\right\}$ *(P.T.U., 2005)* *(iii)* $L\left\{\int_0^t\int_0^t\int_0^t (t\sin t)\,dt\,dt\,dt\right\}$. *(Mumbai, 2006)*

Solution. (*i*) We know that $L(\sin t) = \frac{1}{s^2+1}$

$$L\left(\frac{\sin t}{t}\right) = \int_0^\infty \frac{1}{s^2+1}\,ds = \frac{\pi}{2} - \tan^{-1}s = \cot^{-1}s$$

$$\therefore \quad L\left\{\int_0^t \frac{\sin t}{t}\,dt\right\} = \frac{1}{s}\cot^{-1}s$$

Thus by shifting property, $L\left\{e^{-t}\left(\int_0^t \frac{\sin t}{t}\,dt\right)\right\} = \frac{1}{s+1}\cot^{-1}(s+1)$.

(*ii*) Since $\quad L\left(\frac{\sin t}{t}\right) = \cot^{-1}s$

$$\therefore \quad L\left(e^{-t}\cdot\frac{\sin t}{t}\right) = \cot^{-1}(s+1)$$

and $\quad L\left\{\int_0^t e^{-t}\frac{\sin t}{t}\,dt\right\} = \frac{1}{s}\cot^{-1}(s+1)$

Hence $\quad L\left\{t\cdot\int_0^t e^{-t}\frac{\sin t}{t}\,dt\right\} = -\frac{d}{ds}\left\{\frac{\cot^{-1}(s+1)}{s}\right\}$

$$= -\frac{s\cdot\left[\dfrac{-1}{1+(s+1)^2}\right] - \cot^{-1}(s+1)}{s^2} = \frac{s + (s^2+2s+2)\cot^{-1}(s+1)}{s^2(s^2+2s+2)}$$

(*iii*) Since $\quad L(\sin t) = \frac{1}{s^2+1}$

$$\therefore \quad L(t\sin t) = -\frac{d}{ds}\frac{1}{(s^2+1)} = \frac{2s}{(s^2+1)^2}$$

Thus $\quad L\left\{\int_0^t\int_0^t\int_0^t (t\sin t)\,dt\,.\,dt\,.\,dt\right\} = \frac{1}{s^3}L(t\sin t) = \frac{1}{s^3}\cdot\frac{2s}{(s^2+1)^2} = \frac{2}{s^2(s^2+1)^2}$

21.11 EVALUATION OF INTEGRALS BY LAPLACE TRANSFORMS

Example 21.16. *Evaluate* (i) $\int_0^\infty te^{-3t}\sin t\,dt$ *(V.T.U., 2007)*

(ii) $\int_0^\infty \frac{\sin mt}{t}\,dt$ (iii) $\int_0^\infty e^{-t}\left(\frac{\cos at - \cos bt}{t}\right)dt$ *(Mumbai, 2009)*

(iv) $L\left\{\int_0^t \frac{e^{-t}\sin t}{t}\,dt\right\}$.

Solution. (i) $\int_0^\infty te^{-3t}\sin t\,dt = \int_0^\infty e^{-st}(t\sin t)\,dt$ where $s = 3$

$= L(t\sin t)$, by definition.

$$= (-1)\frac{d}{ds}\left(\frac{1}{s^2+1}\right) = \frac{2s}{(s^2+1)^2} = \frac{2\times 3}{(3^2+1)^2} = \frac{3}{50}.$$

(ii) Since $L(\sin mt) = m/(s^2+m^2) = f(s)$, say.

$\therefore$ Using § 21.10, $L\left(\frac{\sin mt}{t}\right) = \int_s^\infty f(s)\,ds = \int_0^\infty \frac{m\,ds}{s^2+m^2} = \left|\tan^{-1}\frac{s}{m}\right|_s^\infty$

or by Def., $$\int_0^\infty e^{-st}\frac{\sin mt}{t}\,dt = \frac{\pi}{2} - \tan^{-1}\frac{s}{m}$$

Now $\underset{s\to 0}{\text{Lt}}\ \tan^{-1}(s/m) = 0$ if $m > 0$ or π if $m < 0$.

Thus taking limits as $s \to 0$, we get

$$\int_0^\infty \frac{\sin mt}{t}\,dt = \frac{\pi}{2} \text{ if } m > 0 \quad \text{or} \quad -\pi/2 \text{ if } m < 0$$

(iii) We know that $L(\cos at) = \frac{s}{s^2+a^2}$ and $L(\cos bt) = \frac{s}{s^2+b^2}$

$$\therefore \quad L\,\frac{\cos at - \cos bt}{t} = \int_s^\infty \left(\frac{s}{s^2+a^2} - \frac{s}{s^2+b^2}\right)ds$$

$$= \frac{1}{2}\left\{\log\left(\frac{s^2+a^2}{s^2+b^2}\right)\right\}_s^\infty = \frac{1}{2}\log\left(\frac{s^2+b^2}{s^2+a^2}\right)$$

This implies that $\int_0^\infty e^{-st}\left(\frac{\cos at - \cos bt}{t}\right)dt = \frac{1}{2}\log\left(\frac{s^2+b^2}{s^2+a^2}\right)$

Taking $s = 1$, we get $\int_0^\infty \left(e^{-t}\,\frac{\cos at - \cos bt}{t}\right)dt = \frac{1}{2}\log\left(\frac{1+b^2}{1+a^2}\right)$

(iv) Since $$L\left(\frac{\sin t}{t}\right) = \int_s^\infty \frac{ds}{s^2+1} = \tan^{-1} s = \frac{\pi}{2} - \tan^{-1} s = \cot^{-1} s.$$

$\therefore$ $L\left\{e^t\left(\frac{\sin t}{t}\right)\right\} = \cot^{-1}(s-1)$, by shifting property (§ 21.4 *II*).

Thus $L\left[\int_0^t \left\{e^t\left(\frac{\sin t}{t}\right)\right\}dt\right] = \frac{1}{s}\cot^{-1}(s-1)$, by § 21.8.

PROBLEMS 21.3

1. Find $L\left(\int_0^t e^{-t}\cos t\,dt\right)$.

2. Given $L\{2\sqrt{(t/\pi)}\} = 1/s^{3/2}$, show that $L\{1/\sqrt{(\pi t)}\} = 1/\sqrt{s}$. (U.P.T.U., 2005 ; Madras, 2003)

3. Given $L[\sin(\sqrt{t})] \dfrac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4\pi}$, prove that $L\left[\dfrac{\cos(\sqrt{t})}{\sqrt{t}}\right] = \sqrt{\left(\dfrac{\pi}{s}\right)} e^{-1/4\pi}$. (Mumbai, 2009)

Find the Laplace transforms of the following functions :

4. $t\sin^2 t$ (Nagarjuna, 2008)
5. $\sin 2t - 2t\cos 2t$. (Anna, 2003)
6. $t^2\cos at$.
7. $t\sinh at$.
8. $te^{2t}\sin 3t$. (Madras, 2003)
9. $te^{-2t}\sin 4t$. (V.T.U., 2008)
10. $t^2e^{-3t}\sin 2t$. (Madras, 2000 S)
11. $(e^{-at} - e^{-bt})/t$. (Anna, 2005 S)
12. $(\sin t)/t$. (P.T.U., 2010)
13. $\dfrac{(\sin t\sin 5t)}{t}$. (Mumbai, 2008)
14. $(e^{at} - \cos bt)/t$. (U.P.T.U., 2003)
15. $(e^{-t}\sin t)/t$. (V.T.U., 2009 S)
16. $(1 - \cos 3t)/t$. (V.T.U., 2006)
17. $(1 - \cos t)/t^2$. (Hazaribag, 2008)
18. $2^t + \dfrac{\cos 2t - \cos 3t}{t} + t\sin t$. (V.T.U., 2004)

19. Evaluate (*i*) $\displaystyle\int_0^\infty \frac{\cos 6t - \cos 4t}{t}\,dT$ (Mumbai, 2008 ; P.T.U., 2006)

(*ii*) $\displaystyle\int_0^\infty \frac{e^{-\sqrt{2}t}\sinh t\sin t}{t}\,dt$ (Mumbai, 2005) (*iii*) $\displaystyle\int_0^\infty te^{-2t}\sin 3t\,dt$ (V.T.U., 2008)

(*iv*) $\displaystyle\int_0^\infty te^{-t}\sin^4 t\,dt$.

20. Prove that (*i*) $\displaystyle\int_0^\infty \frac{e^{-at} - e^{-bt}}{t}\,dt = \log\frac{b}{a}$. (S.V.T.U., 2009 ; Mumbai, 2007 ; J.N.T.U., 2006)

(*ii*) $\displaystyle\int_0^\infty \frac{e^{-2t}\sinh t}{t}\,dt = \frac{1}{2}\log 3$ (Mumbai, 2008) (*iii*) $\displaystyle\int_0^\infty \frac{e^{-t}\sin t}{t}\,dt = \frac{\pi}{4}$. (V.T.U., 2009 S)

(*iv*) $\displaystyle\int_0^\infty \frac{e^{-t}\sin^2 t}{t}\,dt = \frac{1}{4}\log 5$. (Kurukshetra, 2006)

21. Evaluate (*i*) $L\left(\displaystyle\int_0^t \frac{\sin t}{t}\,dt\right)$ (J.N.T.U., 2005)

(*ii*) $L\left(\displaystyle\int_0^t e^{-t}\cos t\,dt\right)$ (*iii*) $L\displaystyle\int_0^t \frac{e^t\sin t}{t}\,dt$. (P.T.U., 2009 S ; S.V.T.U., 2009 ; Bhopal, 2008)

22. Show that (*i*) $L[t J_0(at)] = \dfrac{s}{(s^2 + a^2)^{3/2}}$ (*ii*) $\displaystyle\int_0^\infty te^{-3t} J_0(4t)\,dt = 3/125$.

21.12 INVERSE TRANSFORMS — METHOD OF PARTIAL FRACTIONS

Having found the Laplace transforms of a few functions, let us now determine the inverse transforms of given functions of s. We have seen that $L\{f(t)\}$ in each case, is a rational algebraic function. Hence *to find the inverse transforms, we first express the given function of s into partial fractions which will, then, be recognizable as one of the following standard forms* :

(1) $\mathbf{L^{-1}\left[\dfrac{1}{s}\right] = 1}$. **(2)** $\mathbf{L^{-1}\left[\dfrac{1}{s - a}\right] = e^{at}}$.